

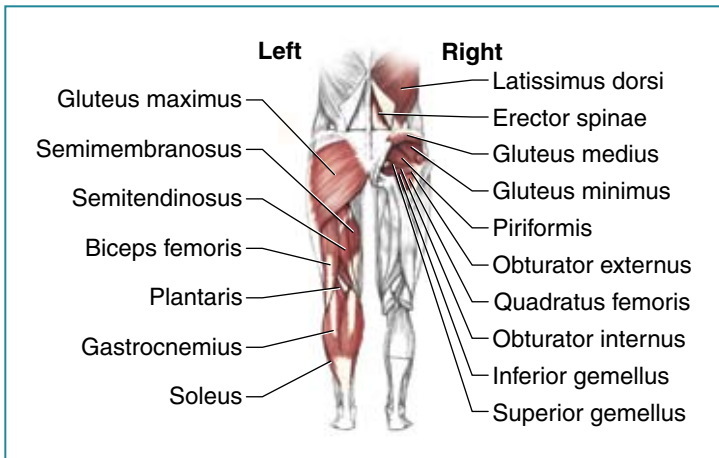
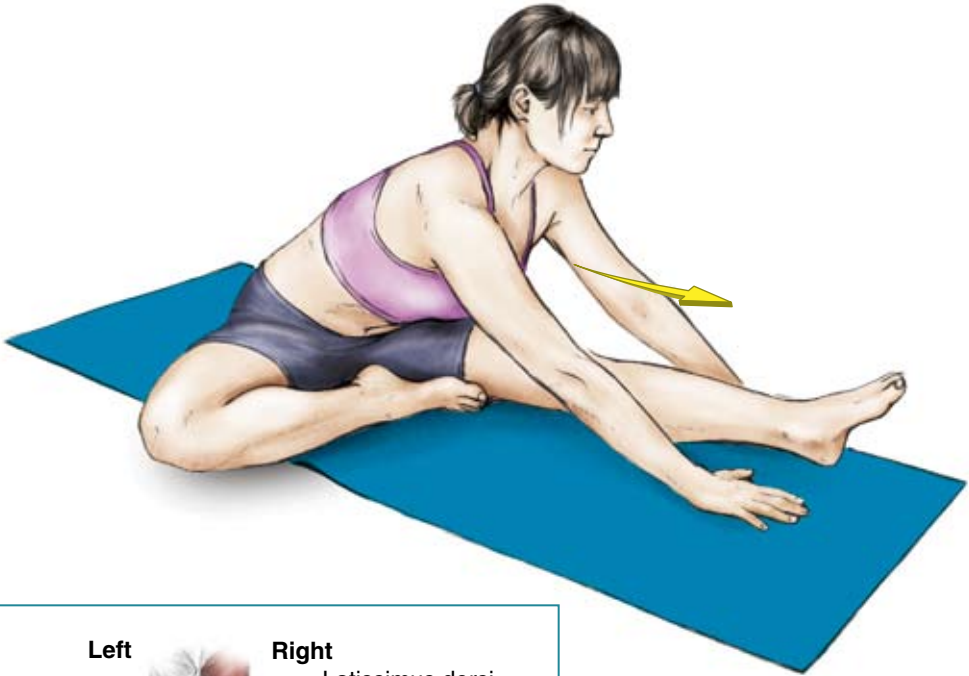
The hip joint is a ball-and-socket joint, which allows a wider range of muscular movements than most other joints in the body. All but two of these muscles (psoas major and piriformis) run between the pelvic bones and the thigh bone (femur); the psoas major and piriformis run between the lower vertebral column and the femur. The muscles that move the hip joint are some of the most massive muscles (adductor magnus and gluteus maximus) in the body as well as some of the smallest (gemellus superior and inferior). The anterior muscles (psoas major, iliacus, rectus femoris, sartorius) flex the hip and are used during walking to swing the leg forward. The posterior muscles (gluteus maximus, biceps femoris, semimembranosus, semitendinosus) provide the backward swing of walking. A group of large muscles (adductor brevis, adductor magnus, adductor longus, gracilis, pectineus) is on the medial (inside) thigh. These muscles keep the legs centered under the body. A group of small muscles (gluteus medius, gluteus minimus, piriformis, gemellus superior, obturator internus, gemellus inferior, obturator externus, quadratus femoris, tensor fasciae latae) is on the lateral (outside) thigh and work to splay the legs to the side. Another group that makes up more than 75 percent of the hip muscles is the external hip rotators (gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, obturator internus, gemellus inferior, obturator externus, quadratus femoris, psoas major, iliacus, rectus femoris, sartorius, adductor brevis, adductor magnus, adductor longus, pectineus). Figures showing these muscles as well as a chart showing specific movements for each muscle are located at the end of the chapter (pages 88-90).

Flexibility has more to do with the overall body function than previously thought. For instance, diminished flexibility is one indicator of an aging body. Decreased physical activity also results in decreased flexibility. As people age and decrease their physical activity, they must keep stretching muscle groups in order to maintain mobility and range of motion in the joints. The hip region is located in the middle of the body, so problems in this area tend to radiate and affect many other parts of the body. You can reduce and even prevent many hip problems by paying more attention to strength and joint flexibility.

Often pain in the hip or buttocks area is associated with poor hip flexibility. This is especially true after running or hiking along steep inclines or declines. Hip pain that occurs one to two days after the activity is due to extensive use of the hip external rotator muscles and is caused by damage to both the muscle and the connective tissues in and around the muscle. Stretching these muscles before and after the activity may help decrease this soreness. In addition, the hip external rotator muscles are the least-stretched muscles of the lower body, probably because these muscle groups are also the most difficult to stretch.

Some of the instructions and illustrations in this chapter are given for one side of the body. Similar but opposite procedures would be used for the opposite side of the body.

Seated Hip External Rotator and Hip Extensor Stretch



Technique

Sit on the floor with the left leg extended straight out in front.

Bend the right knee and place the right foot flat against the left inner thigh, as close as possible to the pelvic area.

Place the hands on the floor next to the thighs.

Bend the trunk over toward the left (straight) knee as far as possible until you start feeling a slight stretch (light pain). Keep the left knee down on the floor if possible as you bend over.

As you bend over, reach out with your arms toward the left foot.

Muscles Stretched

Most-stretched muscles on right side: Gluteus medius and minimus, piriformis, gemellus superior and inferior, obturator externus and internus, quadratus femoris, erector spinae, lower latissimus dorsi.

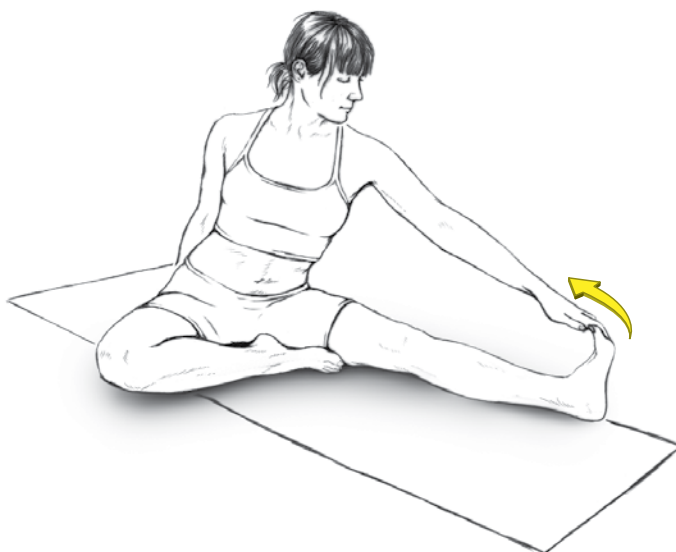
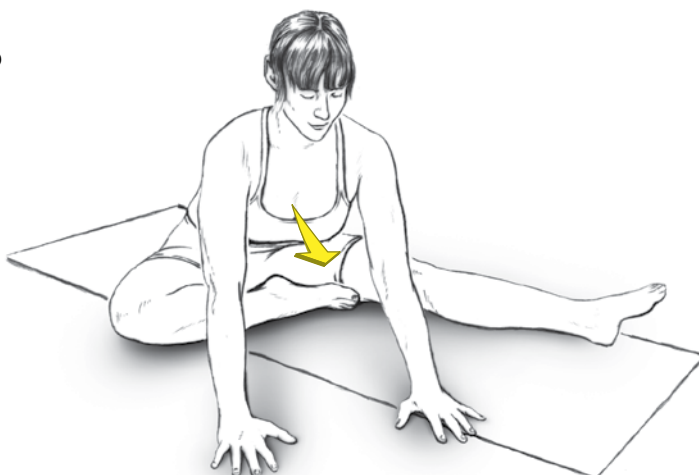
Most-stretched muscles on left side: Semitendinosus, semimembranosus, biceps femoris, gluteus maximus, gastrocnemius.

Lesser-stretched muscles: Soleus, plantaris.

Commentary

Bend the trunk in a forward direction from the hip joint. Keep the trunk as a straight unit; do not let the back curve (see figure below, right). Bending the trunk toward the right knee instead of the left knee reduces the stretch of the most-stretched muscles on the right side of the body and increases the stretch of the most-stretched muscles on the left side of the body.

You can modify this stretch to include the lower-leg muscles (soleus, popliteus, flexor digitorum longus, flexor hallucis longus, posterior tibialis, gastrocnemius, and plantaris). To include these additional muscles, reach out with the left arm, grasp the left foot, and pull the toes slowly toward the knee (dorsiflexed position), as shown in the figure below.



Hip External Rotator Stretch

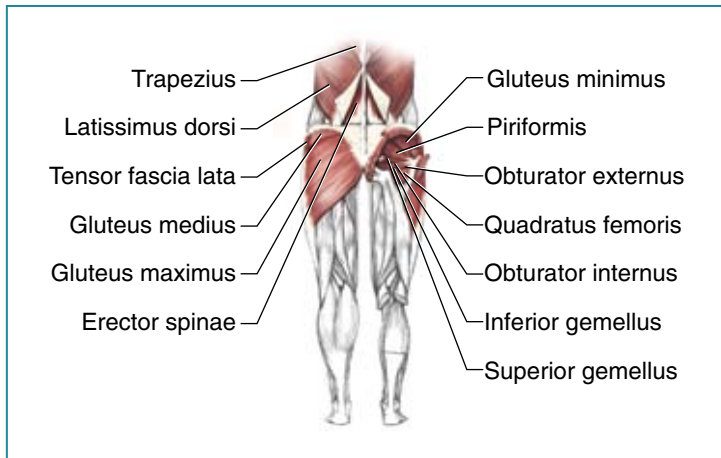


Technique

Stand upright on the left leg with the knee straight; face a support surface (such as a table or beam) that is even with the hips or just a little below the hips.

The right leg is bent at the hip at about a 90-degree angle and rested on the support surface; the outside of the lower right leg rests as flat as possible on the surface. (You can place a towel or pillow under the foot and lower right leg for cushioning.)

Lower the trunk as far as possible toward the right foot, keeping the right knee as flat as possible on the surface.



Muscles Stretched

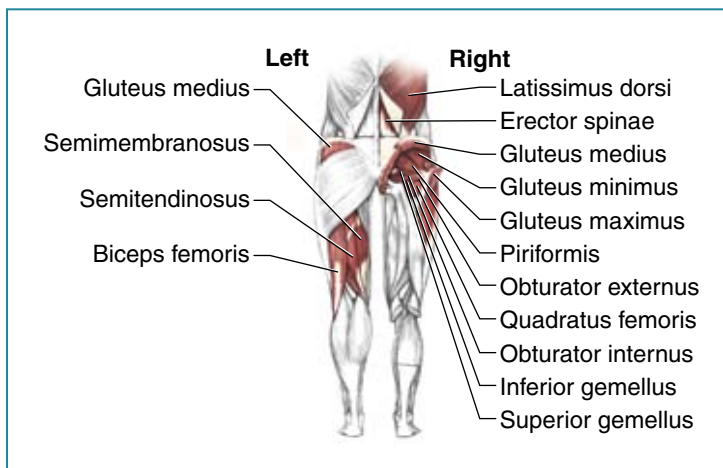
Most-stretched muscles: Gluteus maximus and medius and minimus, right piriformis, right gemellus superior and inferior, right obturator externus and internus, right quadratus femoris, lower erector spinae, left latissimus dorsi.

Lesser-stretched muscles: Right tensor fascia lata, right lower latissimus dorsi, lower trapezius.

Commentary

Lower the trunk forward from the hip joint. Keep the trunk as a straight unit; do not let the back curve. Increasing the height of the table, bench, or other surface by 1 to 2 feet (30 to 61 cm) above the hips will increase the stretch on these muscle groups.

Recumbent Hip External Rotator and Hip Extensor Stretch



Technique

Lie on your back on a comfortable surface.

While outwardly rotating the right leg, bend the right knee and bring the right foot to the body's midline (point the knee laterally).

While keeping the left leg flat, grasp the right knee with the right hand and the right ankle with the left hand. Pull the lower leg as a unit as far as possible toward the chest.

Muscles Stretched

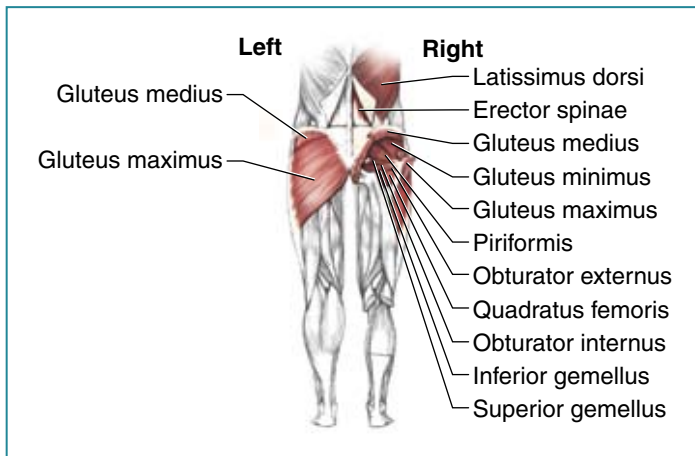
Most-stretched muscles on right side of body: Gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris, lower latissimus dorsi, erector spinae.

Lesser-stretched muscles in left leg: Semitendinosus, semimembranosus, biceps femoris, gluteus medius (if the leg is kept flat on the floor).

Commentary

Bringing the ankle toward the head or even over the head will stretch the aforementioned muscles to the maximum.

Recumbent Hip External Rotator and Hip Extensor Stretch (Crossed Leg)



Technique

Lie on your back on a comfortable surface.

Bend the left leg so that the knee is raised up off the floor while keeping the left foot on the floor.

Bend the right knee and cross the right ankle over and just above the left knee.

Grasp the left leg just under the left knee with both hands.

Pull the left knee along with the bent right knee toward your chest as far as possible until you start feeling a slight stretch (light pain).

Muscles Stretched

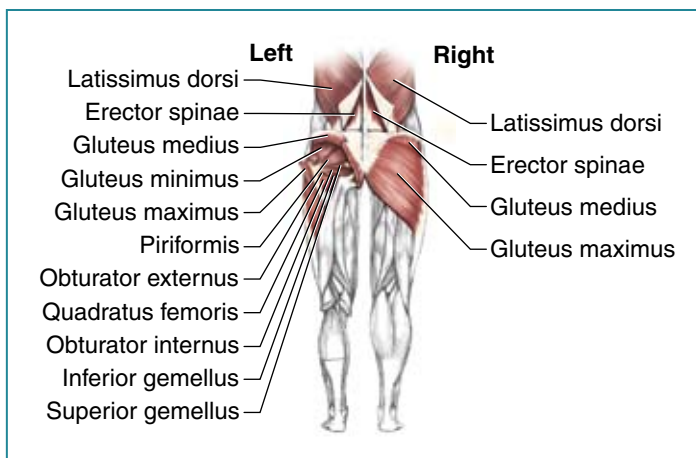
Most-stretched muscles on right side of body: Gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris, lower latissimus dorsi, erector spinae.

Lesser-stretched muscles in left leg: Gluteus maximus, gluteus medius.

Commentary

You can do this stretch while in a sitting position, but it is less effective and more difficult to maintain balance.

Hip External Rotator and Back Extensor Stretch



Technique

Sit on the floor with the right leg extended.

Bend the left leg and place the left foot on the outside of the right knee.

Bend the right arm and position the outside of the right elbow against the outside of the upraised left knee.

Brace the left arm against the floor near the left hip.

Push the right elbow against the left knee, twisting the trunk as far as possible to the left. Maintain enough pressure with the right elbow to keep the left knee in a stable position.

Muscles Stretched

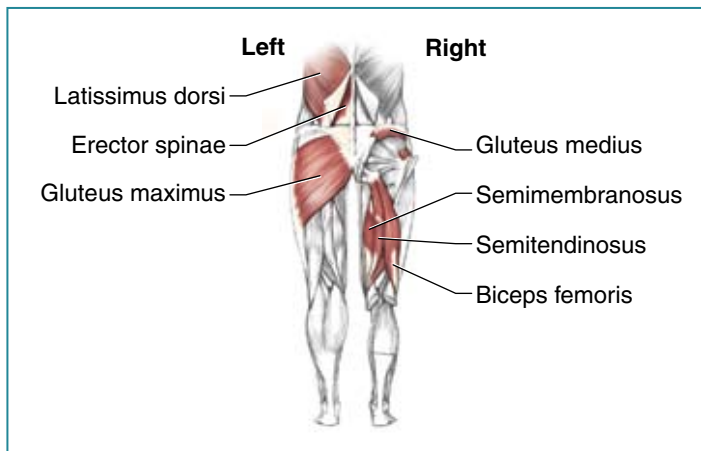
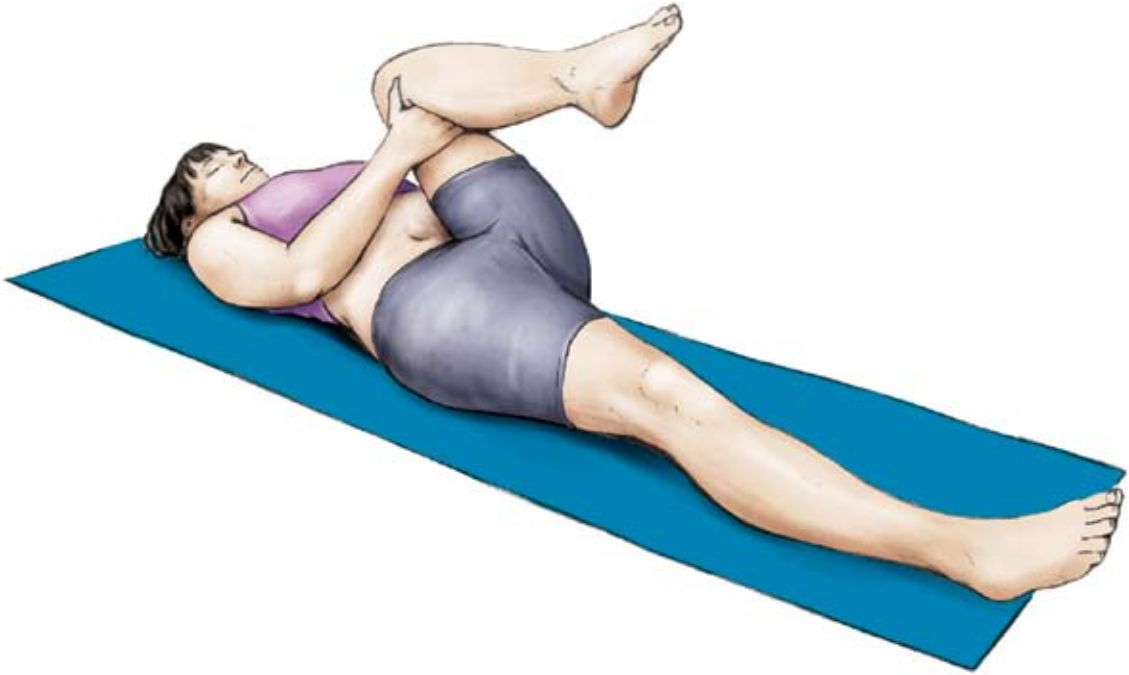
Most-stretched muscles on left side of the body: Gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris, lower latissimus dorsi, erector spinae.

Lesser-stretched muscles on right side of body: Gluteus maximus, gluteus medius, erector spinae, lower latissimus dorsi.

Commentary

Do not arch the back or bend forward at the waist.

Hip Extensor and Back Extensor Stretch



Technique

Lie on your back on a comfortable surface.

Bend the left knee and bring it toward the chest.

While keeping the right leg flat, grasp the left knee with both hands and pull it down toward the chest as far as possible.

Muscles Stretched

Most-stretched muscles: Gluteus maximus, erector spinae, lower latissimus dorsi.

Lesser-stretched muscles in the right leg: Semitendinosus, semimembranosus, biceps femoris, gluteus medius.

Commentary

Bringing the knee toward the armpit instead of the chest will increase the stretch in the muscles. You can do this exercise with both legs simultaneously, but it won't be as effective as when done with each leg separately.

Standing Bent-Knee Hip Adductor Stretch



Technique

Stand upright with the legs more than shoulder-width apart and the left foot turned out.

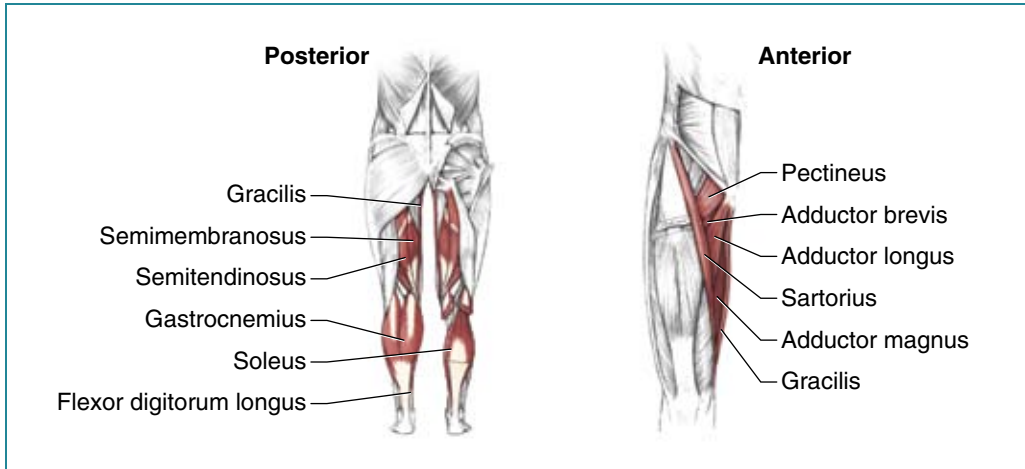
Lower the body (hips) to a half-squatting position, bending the right knee and sliding the left foot outward to the left to keep the left knee straight.

While going down, place the hands on the top of the right knee for support and balance (or hold on to an object for balance).

Muscles Stretched

Most-stretched muscles: Left gracilis, left adductor magnus, left adductor longus, left adductor brevis, left pectineus, middle and lower part of left sartorius, left semitendinosus, left semimembranosus.

Lesser-stretched muscles: Medial side left gastrocnemius and left soleus, left flexor digitorum longus.

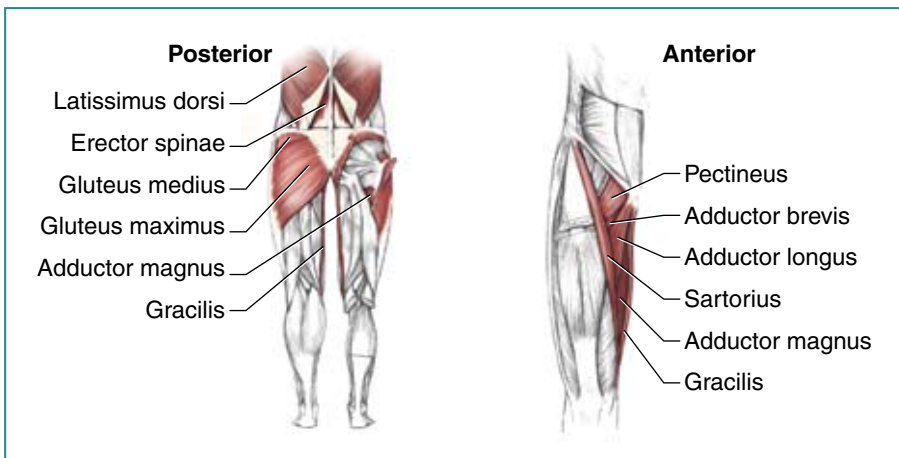
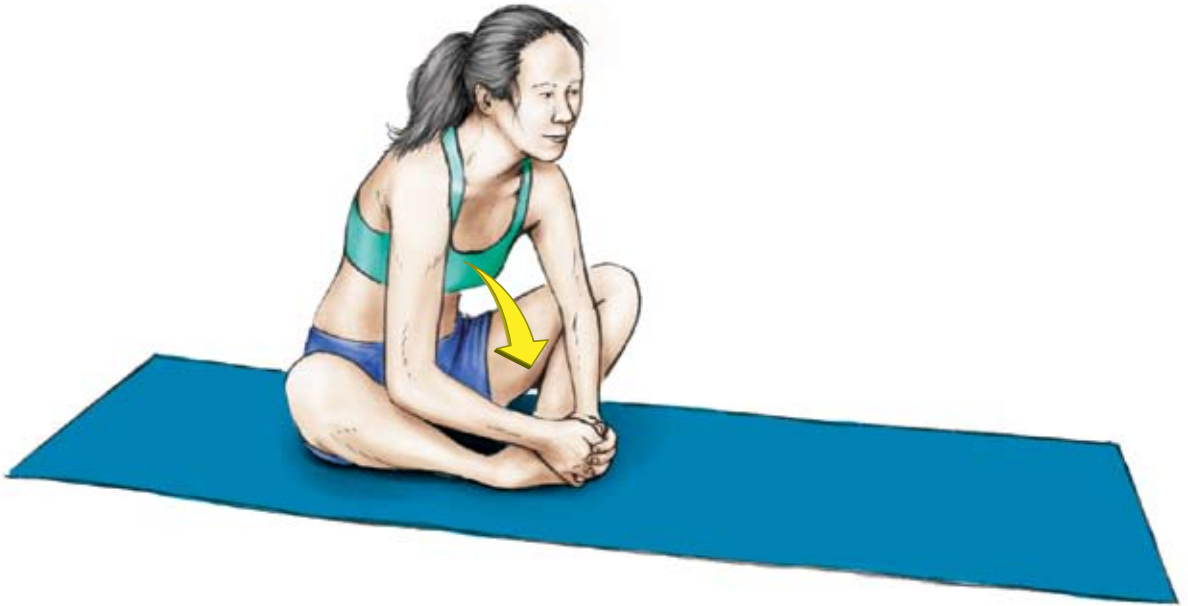


Commentary

Keep the trunk as straight as possible. It is more comfortable to rest the left foot on the inside of the foot. To increase the stretch, bend the trunk to the right and press the right thigh down with both hands at the same time.



Seated Hip Adductor Stretch



Technique

Sit on the floor in the lotus position (knees bent, feet together with the soles touching).

Bring the heels of the feet as close as possible to the buttocks (distance depends on degree of flexibility).

Grasp the feet or just above the ankles with elbows spreading sideways and touching the legs just below the knees.

Bend the trunk over toward the feet, and press the lower part of the thighs and knees down with the elbows while stretching.

Muscles Stretched

Most-stretched muscles: Gracilis, adductor magnus, adductor longus, adductor brevis, pectineus, middle part of sartorius, lower erector spinae, lower latissimus dorsi.

Lesser-stretched muscles: Gluteus maximus, posterior part of gluteus medius.

Commentary

The closer the heels are to the buttocks, the greater the stretch. Placing the heels 1 foot (30 cm) away from the buttocks increases the stretch on the gluteus maximus, gluteus medius, and erector spinae and places the greatest portion of the stretch on the origins of the adductor muscles.

Standing Raised-Leg Hip Adductor Stretch



Technique

Stand upright with weight balanced on the left leg.

Place the right leg on a table, bench, or object that is about even with the height of the hips.

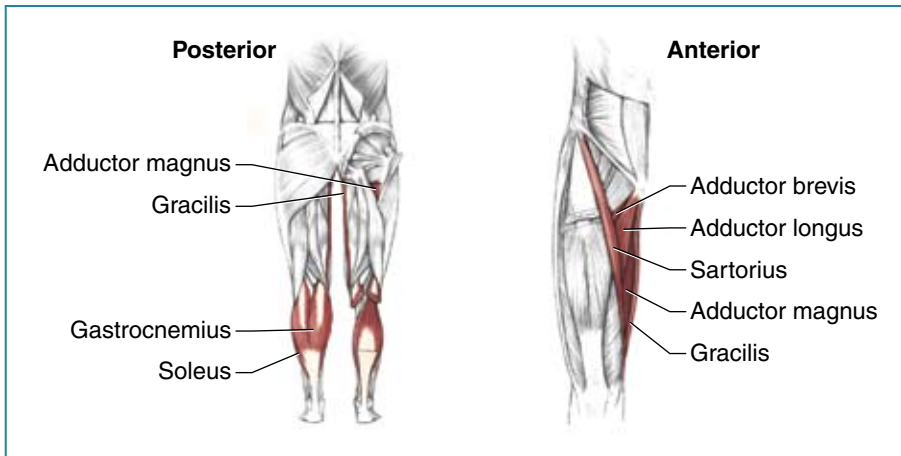
While keeping the right knee straight, rotate the body sideways so that the trunk and the left leg face 90 degrees away from the raised right leg (allow the right leg to rotate so that the right knee points to the side). Point the left knee and toes forward (directly in front of the hips).

Bend the left knee slightly, but keep the right knee straight.

Hang both arms down in front of the left leg with the palms close to the floor.

Alternatively, place the left hand over the left knee and the right hand on the lateral (outer) side of the right knee (as illustrated).

Bend the trunk slightly forward toward the left knee.



Muscles Stretched

Most-stretched muscles: Right gracilis, right adductor magnus, right adductor brevis, right adductor longus, right middle sartorius.

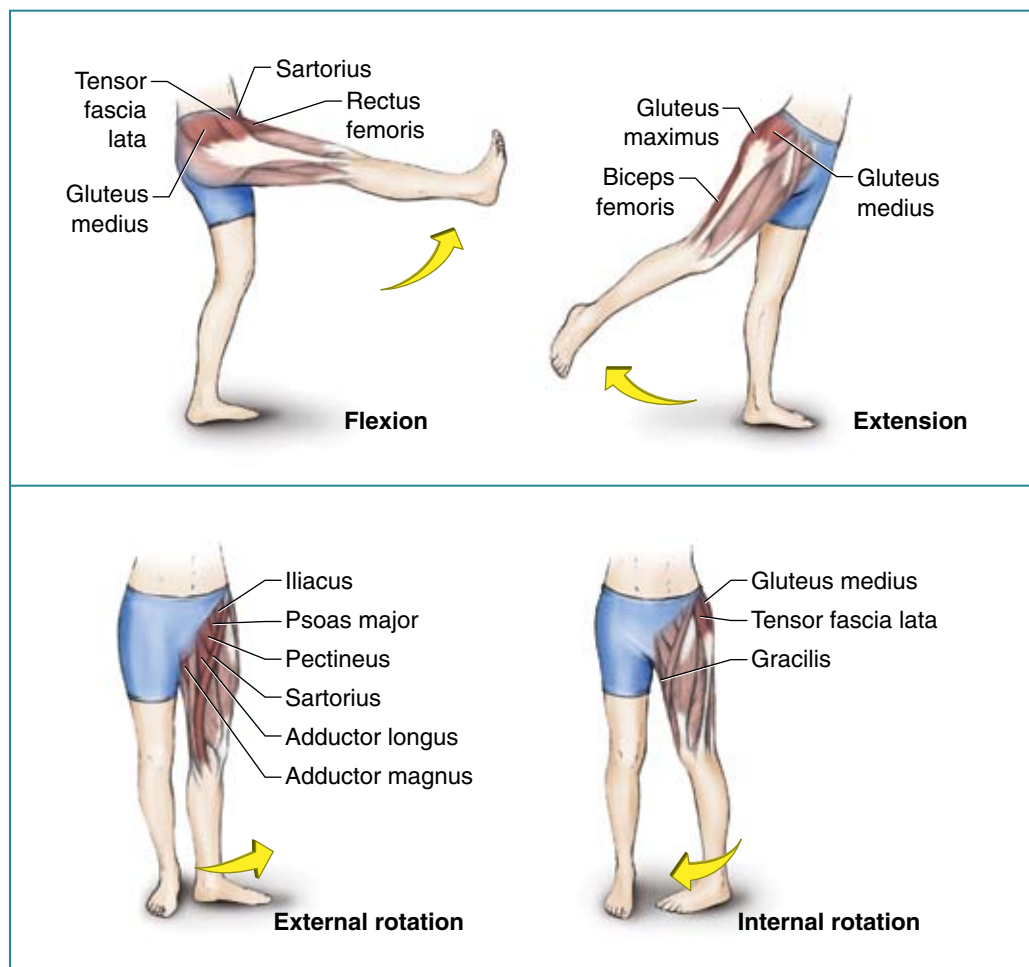
Lesser-stretched muscles: Right medial gastrocnemius, right soleus.

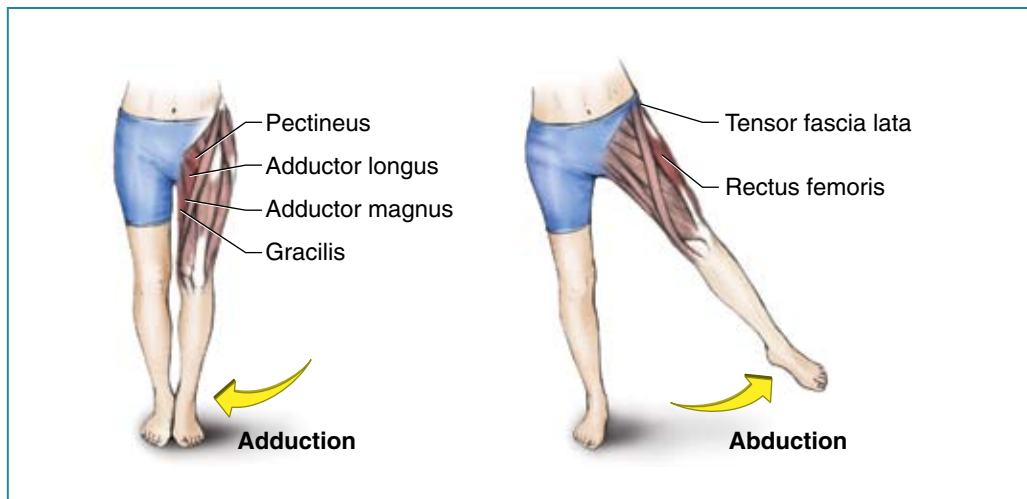
Commentary

Make sure to keep the right knee straight. The alternative hand placement allows for greater stretch especially if you apply pressure with the right hand on the right knee. The stretch is also increased with an increased bend of the left knee.

Hip Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various hip and thigh movements appear in the table on page 90. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the left adductor magnus, you could perform a movement that involves extension, internal rotation, and abduction of the left leg. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (for example, to stretch a very tight adductor magnus, you could start by doing only hip abduction). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.





Muscle	Hip extension	Hip flexion	Hip internal rotation	Hip external rotation	Hip adduction	Hip abduction
Gluteus maximus	✓			✓	✓	✓
Gluteus medius	✓	✓	✓	✓		✓
Gluteus minimus	✓	✓	✓	✓		✓
Biceps femoris	✓			✓		
Semitendinosus	✓		✓			
Semimembranosus	✓		✓			
Rectus femoris		✓				✓
Sartorius		✓		✓		
Pectineus		✓		✓	✓	
Tensor fascia latae		✓	✓			✓
Iliacus		✓		✓		
Psoas major		✓		✓		
Adductor magnus		✓		✓	✓	
Adductor longus		✓		✓	✓	
Adductor brevis		✓		✓	✓	
Gracilis		✓	✓		✓	
Piriformis				✓		✓
Gemellus superior				✓		✓
Obturator internus				✓		
Gemellus inferior				✓		✓
Obturator externus				✓		
Quadratus femoris				✓		